

ESP32 Marauder — Complete Hardware Guide

Firmware: ESP32 Marauder · **Upstream:** [justcallmekoko/ESP32Marauder](#) (GPL-3.0) **Chips:** ESP32, ESP32-S2, ESP32-S3, ESP32-C5 · **Cyber Controller profile:** `marauder` (esptool backend, multi-file offsets, suicide-capable) **This guide:** purchase → build → flash → integrate into Cyber Controller → use → troubleshoot.

1. Overview

ESP32 Marauder is the most widely used ESP32 WiFi/Bluetooth offensive-security firmware. It performs WiFi scanning, packet sniffing, deauthentication, beacon/probe spam, evil-portal/evil-AP, PMKID/EAPOL capture, and BLE scanning/spoofing, with an on-device UI on display boards and a serial CLI everywhere. Cyber Controller flashes it, drives its serial command set, and feeds its scan output into the shared target pool for cross-device coordination.

2. Legal & Safety

Authorized testing only. Deauthentication, beacon spam, and evil-AP transmit on the 2.4 GHz band and are **illegal against networks/people you do not own or have written permission to test** in most jurisdictions (US: CFAA/FCC; EU and others similar). Use only on your own lab gear or under an explicit engagement scope. Cyber Controller labels these as dangerous commands and (unless you disable warnings) confirms before sending. Passive scanning/sniffing may still be regulated where you live — check local law.

3. Hardware & Purchasing

Marauder runs on many ESP32 boards. Pick by how you'll use it:

Tier	Board	Why	Where to buy (search terms)
Best all-in-one	Official Marauder v6/v7 (ESP32-WROOM + 2.4" touch + SD + battery)	Purpose-built, on-device UI, GPS option	Search "justcallmekoko Marauder" on Tindie ; resellers: dbrand Killer , Lonely Binary
Cheap + screen	CYD "Cheap Yellow Display" ESP32-2432S028R (2.8" ILI9341)	~US\$10–15, great starter	AliExpress/Amazon: "ESP32-2432S028R CYD 2.8"
Pocket	M5StickC Plus2 / M5Cardputer (ESP32-S3)	Tiny, built-in screen/keyboard	M5Stack store / Amazon: "M5Cardputer"

Flipper add-on	Flipper Zero WiFi Dev Board (ESP32-S2) or Multi-Board S3	Integrates with Flipper	Flipper official store; "Flipper WiFi dev board"
Bare/no screen	Any ESP32-WROOM-32 DevKit	Cheapest, serial-only via Cyber Controller	AliExpress/Amazon: "ESP32 DevKit V1"
Dual-band	Marauder Mini v3 / Waveshare ESP32-C5	adds 5 GHz	Waveshare: "ESP32-C5 DevKitC"

Accessories: a quality **USB-C/micro-USB data cable** (not charge-only), optional **microSD** (FAT32) for capture logging, optional **GPS module** (NMEA) for wardriving, and an external antenna where the board exposes a u.FL connector. **Avoid** ESP32 clones with the wrong flash size; verify 4MB+.

4. Building / Assembly

- **Pre-built boards (CYD, M5, official Marauder):** no assembly — just a data cable.
- **DevKit + screen DIY:** wire an SPI display per the upstream wiring guide; most users skip this and use a CYD instead.
- **Drivers:** install the USB-serial driver for your board's chip — **CP210x** (most DevKits/official), **CH340** (cheap clones), or native USB (S2/S3). Without it, no COM port appears.
- **Per-board gotchas:** CYD comes in display variants (2432S028 vs Guition 2.4"/3.5") — pick the matching variant in Cyber Controller or the screen stays blank. ESP32-C5 uses a **0x2000** bootloader offset (Cyber Controller handles this automatically).

5. Flashing & First Run (via Cyber Controller)

1. Connect the board by USB; open Cyber Controller → **Flash** tab.
2. **Port:** pick the board's COM/tty (click *Refresh* if missing → driver issue).
3. **Firmware Profile:** ESP32 Marauder.
4. **Board / variant:** choose your exact board (e.g. *CYD 2.8"*, *M5Cardputer*, *Generic ESP32*). "Auto" uses the chip default and may be wrong for display boards — if the screen stays blank, re-flash with the matching variant.
5. Click **Flash**. Cyber Controller auto-detects the chip, fetches the latest release, and writes the bootloader/partitions/app at the correct offsets. First boot shows the Marauder UI (on screen boards) or prints a banner on the serial terminal.

6. Integrate into Cyber Controller

- **Profile:** marauder (esptool, multi-file: bootloader@0x1000 / partitions@0x8000 / app@0x10000; the app pulls boot files from the upstream FlashFiles tree per chip).
- **Control:** open the **Devices** tab → select the port → set **Firmware = ESP32 Marauder** (Pro mode; Simple mode auto-uses it) → **Connect**. The protocol-aware parser understands Marauder output, so APs/ clients populate the **Targets** pool and the persistent terminal colorizes per device.
- **Command set:** the per-firmware palette offers Marauder commands (`scanap`, `sniffraw`, `attack -t` `deauth`, `list -a`, `select -a <n>`, `stopscan`, etc.). Dangerous verbs are confirmed first.

- **Cross-Comm:** a scan on the Marauder feeds `target.added` events; auto-routing rules can fire a command on another connected device. **Dead Man's Switch** is supported for this profile.
- **Backup first:** use **Backup** in the Flash tab to dump the current flash before overwriting.

7. Usage (end-to-end)

1. **Scan:** Devices tab → scanap → watch APs appear in **Targets**.
2. **Select a target** in Targets → right-click → action (e.g. deauth) — or send `select -a <n>` then attack `-t deauth` in the terminal (authorized targets only).
3. **Capture:** `sniffraw` / PMKID modes; with a microSD the device logs `.pcap/.log`.
4. **Wardrive:** pair a GPS module and use the **Wardrive** tab to export WiGLE CSV (lawful, owner-authorized).
5. **Stop:** `stopscan` / stop the attack; Disconnect to free the port.

8. Troubleshooting

- **No COM port:** install CP210x/CH340 driver; try another (data) cable; on Linux add your user to `dialout` and replug.
- **Blank screen after flash:** wrong board variant — re-flash selecting the exact display board.
- **Flash fails / "Failed to connect":** hold **BOOT** while plugging in (or during connect), lower the flash baud in Settings, close any serial monitor holding the port.
- **Boot-loops / brick:** use **Erase Flash** then re-flash; verify the board is genuinely 4MB+.
- **C5/dual-band oddities:** ensure the C5 variant is selected (the `0x2000` bootloader offset is required — Cyber Controller sets it automatically).
- **Verify failed:** re-run flash with verify on; a bad cable/clone flash chip is the usual cause.

9. Sources

- Upstream: <https://github.com/justcallmekoko/ESP32Marauder> (README, wiki, FlashFiles, releases).
- Cyber Controller profile: `src/config/profiles/marauder.json` (boards, offsets, command protocol).
- Hardware: official Marauder (Tindie/dbrand/Lonely Binary), CYD (ESP32-2432S028R), M5Stack, Flipper store.
- Verify current board availability + prices at purchase time; vendor links change.